

Vansh Malhotra

AI Engineer | Multi-Agent Systems | LLM Infrastructure | Microservices

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SUMMARY

AI Engineer who founded and scaled eInfoChips' AI Studio initiative from a solo proof-of-concept into a 9-11 member team, productized company-wide as **NomAIzo™ AI Studio**. Builds multi-agent LLM systems, RAG infrastructure, and adversarial LLM security testing across production platforms.

EDUCATION

NEW LJJET (Affiliated with GTU) | GPA: 3.7/4.0
B.E. in Information Technology

Nov 2020 – May 2024
Ahmedabad, India

TECHNICAL SKILLS

Languages & Backend Python, C/C++, FastAPI, Uvicorn, Pydantic, SQLAlchemy, OAuth2/JWT, Nginx, Kong API Gateway
AI/LLM Engineering Agno, CrewAI, LangGraph, LangChain, RAG Systems, Multi-Agent Systems, Fine Tuning
Data & Infrastructure PostgreSQL, PgVector, MongoDB, Redis, RabbitMQ, Docker, Kubernetes (AKS), Helm, Azure DevOps
Evaluation & Observability DeepEval, Garak, Langfuse, Wiz

EXPERIENCE

AI Engineer

July 2024 – Present

eInfoChips (An Arrow Company)

Ahmedabad, India

- Founded **AI Studio** as a solo proof-of-concept and grew it into a **9-11 member team**, mentoring engineers and contributing to key technical and architectural decisions.
- Product was productized and launched company-wide as **NomAIzo™ AI Studio**, an AI-driven testing accelerator now marketed externally by eInfoChips, generating test cases for embedded, desktop, mobile, and web/app systems at **~60% accuracy** against manual baselines.
- Extended automation coverage across web, desktop, and mobile through **Playwright**- and **Appium**-based AI test execution, reducing manual scripting effort across QA teams.
- Built **CyberStudio**, a multi-agent AI platform automating EU Cyber Resilience Act and RED/EN 18031 compliance gap analysis, achieving **~70% analysis accuracy** for internal cybersecurity teams.
- Built **Assessment Studio**, an internal exam platform with slot booking, tab-switch proctoring, and auto-submit safeguards, used by **~500 employees** across Cyber Security, QA, and DevOps divisions.
- Established LLM observability and adversarial testing standards across all platforms using **Langfuse** and **Garak**, achieving **90% security accuracy** across 28 attack probes.
- Owned platform security end-to-end: Fernet-encrypted API key storage, Redis-based JWT blacklisting, bcrypt password hashing, Kong rate limiting, integrated **Wiz** for continuous code and container vulnerability scanning, and closed a JWT privilege-escalation finding from a VAPT audit.

AI Engineer Intern

Jan 2024 – Jul 2024

Seaftux Technologies

Ahmedabad, India

- Built RESTful backend services for an e-commerce platform using FastAPI — covering product catalogue, cart management, and async order processing.

PROJECTS

AI Studio — Enterprise AI Platform for Autonomous QA & Document Intelligence (*eInfoChips*)

2024 – Present

- Architected a **14-microservice** platform on FastAPI with event-driven communication via RabbitMQ and REST, backed by PostgreSQL, PgVector, and Redis, with schema migrations managed through Liquibase.
- Built a RAG pipeline — Docling-based OCR extraction → chunking → PgVector embedding → similarity search → context-augmented LLM response.
- Designed a multi-agent chat system on the **Agno** framework using a supervisor-routing Team architecture, plus **20+ specialized LLM agents** across 5 services for user story generation, test case generation, and test report analysis.
- Built AI-driven test automation across three surfaces — **Playwright** for web/desktop, **Appium** for mobile — using locator-based script generation to autonomously convert recorded interactions into executable test scripts.
- Implemented layered AI safety guardrails — regex filtering, OpenAI Moderation API, and prompt injection detection run concurrently before every LLM call, with Pydantic-enforced structured outputs.
- Collaborated with the DevOps team on Helm-based AKS deployment (30 templates) and ADO Pipelines CI/CD across dev/QA environments.

CyberStudio — Multi-Agent Compliance Gap Analysis Platform (*eInfoChips*)

2025 – Present

- Engineered **6 specialized GPT-4o agents** (Agno framework) automating compliance assessment against **33 EU CRA clauses** and **14 EN 18031** security categories, producing evidence-backed, structured Pydantic outputs.
- Built a RAG pipeline with dual, physically isolated PgVector knowledge bases (CRA and RED) using hybrid semantic + keyword search and content-hash deduplication.
- Designed a 3-phase human-in-the-loop workflow with Redis-backed state, delivering reports across 4 formats (HTML, Word, Excel, PDF) with full Langfuse observability.

Assessment Studio — Internal Exam & Certification Platform (*eInfoChips*)

2026

- Built a full exam lifecycle platform — slot booking, timed delivery, auto-grading, and certificate generation — used by **~500 employees** across Cyber Security, QA, and DevOps divisions.
- Implemented browser-level proctoring with tab-switch/window-blur detection, real-time warnings, and automatic submission on repeated violations.
- Used Redis for exam session state and timing enforcement, with PostgreSQL for question banks, attempts, and results across concurrent test-takers, containerized via Docker.

AWARDS

- **Innovation Award, eInfoChips (2026)** — for founding and scaling AI Studio from a solo proof-of-concept into a company-wide productized platform (NomAIzo™ AI Studio).

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional